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| **Fiscal Year:** | FY 2010 | **Task Last Updated:** | FY 12/06/2011 |
| **PI Name:** | Mourad, Pierre D Ph.D. | | |
| **Project Title:** | Prototype Testing for Non-Invasive Determination of Intracranial Pressure | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Smart Medical Systems and Technology Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2004 NSBRI NNH04ZUU003N Human Health in Space |
| **Start Date:** | 06/01/2005 | **End Date:** | 05/31/2010 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 5/31/2010 (from 5/31/2009) per NSBRI (1/2009) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Kliot, Michel ( University of Washington ) | | |
| **Grant/Contract No.:** | NCC 9-58-SMS00701 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Intracranial pressure (ICP) is a vital determinant of brain function whose monitoring and treatment are central to the successful management of patients with a variety of brain diseases and disorders. There are also indications that changes in cerebral autoregulation - one of the primary physiological mechanisms for maintaining normal ICP, and detectable via alteration of blood flow into the brain - can be altered during space flight in ways associated with space motion sickness and orthostatic intolerance. Currently, ICP is measured by drilling a hole into the skull and placing a device into one of several places within the cranium. This is clearly not an option available to astronauts in space, who often develop brain edema and its associated increase in ICP. Therefore, a simple, non-invasive method of monitoring ICP (nICP) would allow measurement of this critical parameter in astronauts. We have developed several algorithms for nICP based on several different and novel analysis methods, most of which using data derived from standard diagnostic ultrasound of | | |

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| **Task Description:** | the brain and invasively and continually monitored arterial blood pressure (ABP), and have shown it feasible. We have also assembled and tested a device that allows collection of the data necessary to implement in real time, and to further refine this algorithm.<br>Specific Aim #1: to test the feasibility of our approach to determining nICP on humans at several hospitals using invasive ABP (iABP), by collecting additional patient data to be used for refining or the existing algorithm.<br><br>Specific Aim #2: to perform the same feasibility tests and algorithm refinement with non-invasive ABP (nABP).<br><br>Therefore, the work proposed here offers to perform steps necessary to bringing to space a non-invasive means of determining intracranial pressure. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | 1) Non-invasively predicting intracranial pressure (nICP): the patent literature and, to a lesser extent, the scientific literature, are littered with failed attempts to determine intracranial pressure in a non-invasive fashion. These considerable efforts have been motivated by the vital role played by this medical variable in the management of individuals suffering from head trauma, subarachnoid hemorrhage and other sources of intracranial bleeding, as well as brain tumors, hydrocephalus, edema, among other disorders. We have demonstrated in our work for this proposal as well as in our on-going efforts for this proposal several novel means of predicting nICP that has the best chance to date to capture this critical medical parameter whose current means of measurement requires that a neurosurgeon drill a hole through the skull.<br>2) In work with our industrial collaborators we succeeded in automating transcranial Doppler process for the purposes of deriving medically useful information from measurements of blood flow rate the middle cerebral artery and arterial blood pressure. Given such a system, this would allow researchers and clinicians to easily and autonomously measure the rate of blood flow into the brain - of intrinsic value and a critical parameter in our approach to the non-invasive determination of intracranial pressure. This system will also allow quantification of the existence and extent of emboli transport into the brain, and the quality of cerebral autoregulation. The monitoring of these measurements plays an important role in guiding the medical and/or surgical care of patients with a variety of neurological disorders (as listed above). |
| **Task Progress:** | We have finished using our unique computational and logistical infrastructure necessary to most effectively pursue our human-based studies on the non-invasive determination of intracranial pressure. To summarize our effort in support of the logistical infrastructure, we established and used at four different pre-clinical trial sites (those associated with the University of Washington, University of Pennsylvania, Barrow Neurological Institute, and Columbia University) web-based data entry system that allows all of our clinical sites to transmit the human data they collect in a HIPPA compliant way to the University of Washington for analysis. We collected a total of 159 data sets from patients for our studies and have processed that data.<br>Our on-going modeling efforts for determining ICP based upon novel statistical measures remain consistent with our initial estimates of the complexity of the particular mathematical models we are using to relate in an empirical fashion readily observable human physiological data to intracranial pressure. To our regret we could not identify a non-invasive ABP device to use to support our algorithm. We are finalizing one document (for publication, along with an associated patent application) that describes our approach to nICP and nCPP that uses invasive arterial blood pressure devices. We will resubmit a second publication that tests another algorithm for CPP, published by Czosnyka and colleagues. In future studies we intend to explore new nABP devices, to embody the predictive algorithms into our automated transcranial Doppler (aTCD) device, and to field trial both the aTCD alone and with the embodied algorithms working with our industrial partner, PhysioSonics. |
| **Bibliography Type:** | Description: (Last Updated: 11/21/2013) |
| **Abstracts for Journals and Proceedings** | Mourad PD, Marzban C, Kliot M. "Towards predicting intracranial pressure using transcranial Doppler and arterial blood pressure data." 157th annual meeting of the Acoustical Society of America, Portland Oregon, May 18-22, 2009. Journal of the Acoustical Society of America. 2009 Apr;125(4 Pt 2):2514. http://dx.doi.org/10.1121/1.4783446 , Apr-2009 |
| **Articles in Peer-reviewed Journals** | Mourad PD, Illian PR, Kliot M, Ollos RJ, Marzban C, Moore A. "Evaluation of a TCD-based model for estimating cerebral perfusion pressure." Neurosurgery. In press, 2011. , Dec-2011 |
| **Articles in Peer-reviewed Journals** | Marzban C, Illian PR, Morison D, Mourad PD. "A double-gaussian, percentile-based method for estimating maximum blood flow velocity." J Ultrasound Med. 2013 Nov;32(11):1913-20. http://dx.doi.org/10.7863/ultra.32.11.1913 ; PubMed PMID: 24154894 , Nov-2013 |